

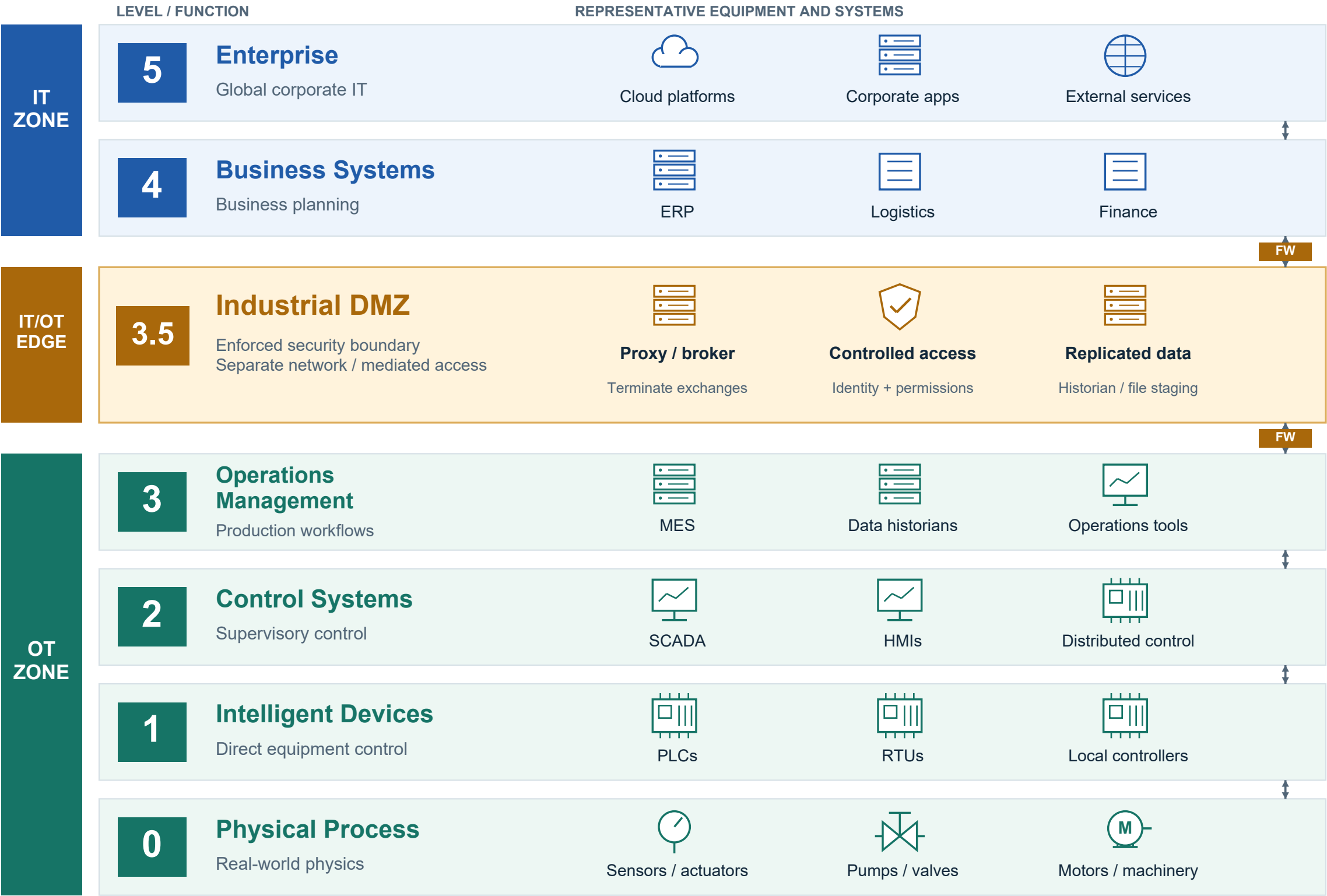
THE PURDUE MODEL

IT / OT segmentation | Six functional levels + an industrial DMZ security boundary

REFERENCE ARCHITECTURE

11 x 17 in / Landscape

Cybersecurity adaptation of the Purdue Enterprise Reference Architecture (PERA).



01 Network segmentation

Mediate approved IT/OT exchanges through the IDMZ. Firewall rules and access controls enforce the boundary.



No direct corporate-to-control access.

02 Limit the blast radius

A compromised corporate email account should not provide a direct path to PLC logic or process setpoints. Enforced boundaries help limit lateral movement.

Risk reduction depends on configuration, monitoring and controls within OT.

03 IT/OT convergence

IIoT and cloud services can cross the traditional layers. Review every path; uncontrolled bypasses create exposure.



ZERO TRUST + SEGMENTATION

Verify identity; grant least privilege; monitor sessions and data flows. Apply controls compatible with OT safety, availability and device capabilities.

↔ Logical exchange; only approved flows are permitted.

----- Illustrative authenticated IIoT / cloud data path.

FW = firewall | Level 3.5 is a security extension to the model.

Conceptual model, not a deployment topology. Systems may span levels; process safety requires its own engineered protections.

Sources: NIST SP 800-82 Rev. 3, Guide to OT Security

Cisco CPwE Industrial DMZ Overview

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